

NET 731 Practical-Week 2

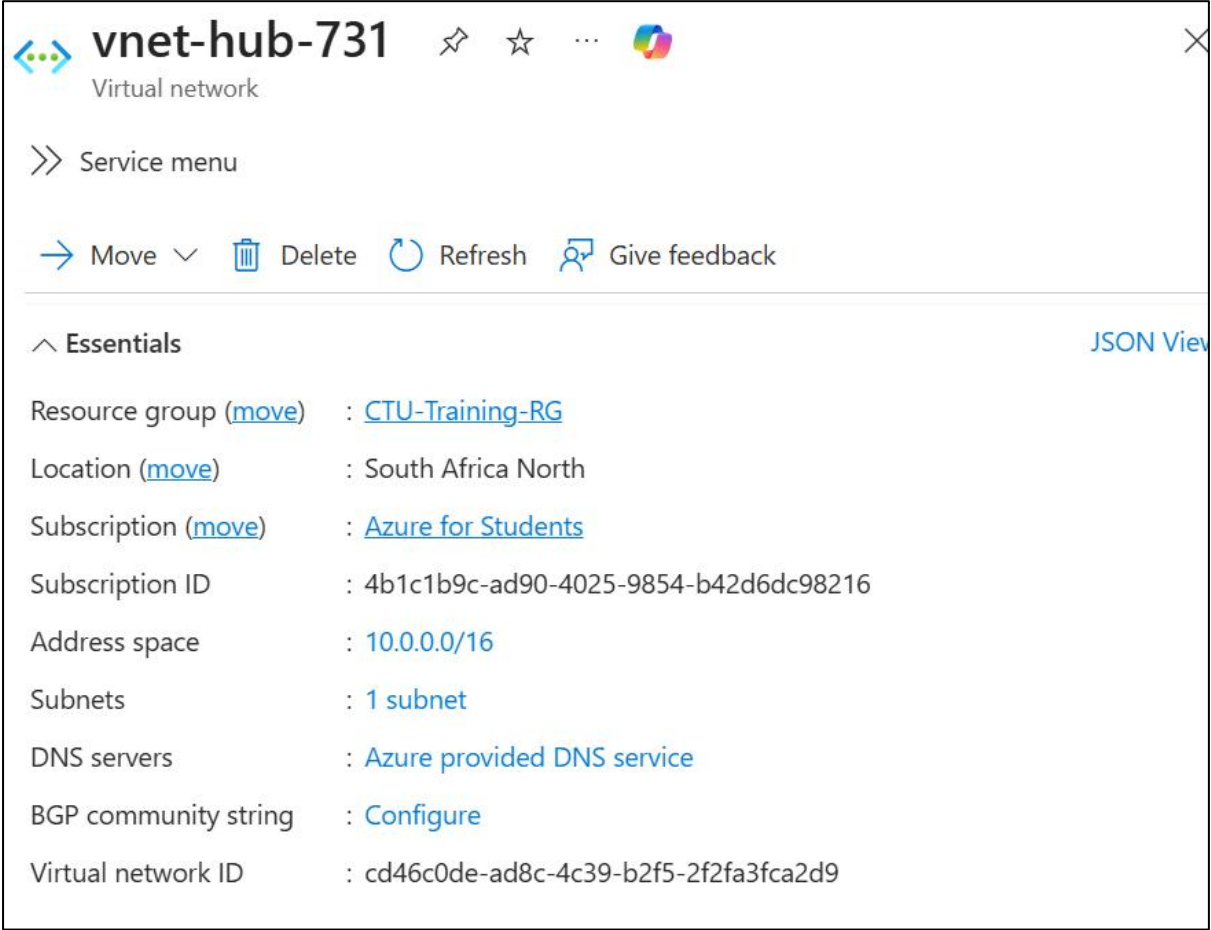
Project Overview

This capstone project simulates the role of an Azure Administrator responsible for designing, deploying, securing, and monitoring a cloud infrastructure using Microsoft Azure. Students must implement Azure services while ensuring the total cost remains within the \$100 Azure for Students credit.

Week 2 (13–19 March) – Network Design (10 Marks)

- Create Virtual Network
- Configure address space and subnets
- Attach Network Security Group to subnets
- Upload network design documentation to GitHub

Create VNet



The screenshot displays the Azure portal interface for a virtual network named 'vnet-hub-731'. The page title is 'vnet-hub-731' with the subtitle 'Virtual network'. Below the title, there is a 'Service menu' section with options: 'Move', 'Delete', 'Refresh', and 'Give feedback'. The main content area is titled 'Essentials' and lists various configuration details for the virtual network. A 'JSON View' link is visible in the top right corner of the Essentials section.

Property	Value
Resource group (move)	: CTU-Training-RG
Location (move)	: South Africa North
Subscription (move)	: Azure for Students
Subscription ID	: 4b1c1b9c-ad90-4025-9854-b42d6dc98216
Address space	: 10.0.0.0/16
Subnets	: 1 subnet
DNS servers	: Azure provided DNS service
BGP community string	: Configure
Virtual network ID	: cd46c0de-ad8c-4c39-b2f5-2f2fa3fca2d9

Configure address space and subnets

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ	Default
Name * ⓘ	Frontend-subnet
IPv4	
Include an IPv4 address space	☒
IPv4 address range ⓘ	10.0.0.0/16 10.0.0.0 - 10.0.255.255
Starting address * ⓘ	10.0.1.0
Size ⓘ	/24 (256 addresses)
Subnet address range ⓘ	10.0.1.0 - 10.0.1.255

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ	Default
Name * ⓘ	Backend-Subnet
IPv4	
Include an IPv4 address space	☒
IPv4 address range ⓘ	10.0.0.0/16 10.0.0.0 - 10.0.255.255
Starting address * ⓘ	10.0.2.0
Size ⓘ	/24 (256 addresses)
Subnet address range ⓘ	10.0.2.0 - 10.0.2.255

IPv6

[Give feedback](#)

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ	Default
Name * ⓘ	Database-Subnet
IPv4	
Include an IPv4 address space	☒
IPv4 address range ⓘ	10.0.0.0/16 10.0.0.0 - 10.0.255.255
Starting address * ⓘ	10.0.3.0
Size ⓘ	/24 (256 addresses)
Subnet address range ⓘ	10.0.3.0 - 10.0.3.255

IPv6

[Give feedback](#)

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ	Default
Name * ⓘ	Management-Subnet
IPv4	
Include an IPv4 address space	☒
IPv4 address range ⓘ	10.0.0.0/16 10.0.0.0 - 10.0.255.255
Starting address * ⓘ	10.0.4.0
Size ⓘ	/24 (256 addresses)
Subnet address range ⓘ	10.0.4.0 - 10.0.4.255

IPv6

[Give feedback](#)

Create Network Security Group (NSG)

Create network security group

...

×

Basics

Tags

Review + create

Project details

Subscription *

Azure for Students

Resource group *

CTU-Training-RG

Create new

Instance details

Name *

NSG-WEB

Region *

South Africa North

 Add inbound security rule

×

NSG-WEB

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

☒ Allow

☐ Deny

Priority * ⓘ

100

Name *

Allow-Internet-HTTP

Description

Add

Cancel

Give feedback

Attach NSG to Subnets

All services > Resource Manager | Resource groups > CTU-T

vnet-hub-731 | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

+ Subnet

Refresh

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐ Name ↑

☒ Frontend-S...

☐ Backend-S...

☐ Database-S...

☐ Manageme...

Showing 4 subnets

Edit subnet

Simplifying internet access for virtual machines by using a network address translation gateway. First subnet came using a network security group. [Learn more](#)

NAT gateway

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

Network security group

Route table

Service Endpoints

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services

Select a service endpoint

Remove service endpoint

Subnet Delegation

Save

Cancel

[Give feedback](#)

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	Backend-S...	10.0.2.0/24	-	251	-	-	-		
☐	Database-S...	10.0.3.0/24	-	251	-	-	-		
☐	Manageme...	10.0.4.0/24	-	251	-	-	-		
☐	Frontend-S...	10.0.1.0/24	-	251	-	NSG-WEB	-		

vnet-hub-731 | Subnets

Virtual network

Search

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐ Name ↑

☐ Frontend-S...

☐ Backend-S...

☐ Database-S...

☐ Manageme...

Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
Frontend-S...	10.0.1.0/24	-	251	-	NSG-WEB	-		
Backend-S...	10.0.2.0/24	-	251	-	NSG-WEB	-		
Database-S...	10.0.3.0/24	-	251	-	NSG-WEB	-		
Manageme...	10.0.4.0/24	-	251	-	NSG-WEB	-		

Successfully saved subnet 'Management-Subnet'.